Journal of
PERIODONTAL RESEARCH

J Periodont Res 2014; 49: 801–809
All rights reserved

© 2014 John Wiley & Sons A/S.
Published by John Wiley & Sons Ltd

JOURNAL OF PERIODONTAL RESEARCH
doi:10.1111/jre.12166

Inhibitory effects of sword bean extract on alveolar bone resorption induced in rats by *Porphyromonas gingivalis* infection

Y. Nakatsuka[1], T. Nagasawa[1], Y. Yumoto[1], F. Nakazawa[2], Y. Furuichi[1]

[1]Division of Periodontology and Endodontology, Department of Oral Rehabilitation, School of Dentistry, Health Sciences University of Hokkaido, Hokkaido, Japan and [2]Division of Microbiology, Department of Oral Biology, School of Dentistry, Health Sciences University of Hokkaido, Hokkaido, Japan

Nakatsuka Y, Nagasawa T, Yumoto Y, Nakazawa F, Furuichi Y. Inhibitory effects of sword bean extract on alveolar bone resorption induced in rats by Porphyromonas gingivalis *infection. J Periodont Res 2014; 49: 801–809. © 2014 John Wiley & Sons A/S. Published by John Wiley & Sons Ltd*

Background: The domesticated legume, *Canavalia gladiata* (commonly called the sword bean), is known to contain canavanine. The fruit is used in Chinese and Japanese herbal medicine for treating the discharge of pus, but its pharmacological mechanisms are still unclear.

Objectives: This study examined the effect of sword bean extract (SBE) on (i) oral bacteria and human oral epithelial cells *in vitro*, and (ii) the initiation and progression of experimental *Porphyromonas gingivalis*-induced alveolar bone resorption in rats.

Material and Methods: A high-performance liquid chromatography/ultraviolet method was applied to quantitate canavanine in SBE. By assessing oral bacterial growth, we estimated the minimum inhibitory concentration and minimum bactericidal concentration of SBE, canavanine, chlorhexidine gluconate (CHX) solution. The cytotoxicity of SBE, canavanine, CHX, leupeptin and cystatin for KB cells was determined using a trypan blue assay. The effects of SBE, canavanine, leupeptin and cystatin on Arg-gingipain (Rgp) and Lys-gingipain (Kgp) were evaluated by colorimetric assay using synthetic substrates. To examine its effects on *P. gingivalis*-associated periodontal tissue breakdown, SBE was orally administered to *P. gingivalis*-infected rats.

Result: Sword bean extract contained 6.4% canavanine. SBE and canavanine inhibited the growth of *P. gingivalis* and *Fusobacterium nucleatum*. The cytotoxicity of SBE, canavanine and cystatin on KB cells was significantly lower than that of CHX. Inhibition of Rgp with SBE was comparable to that with leupeptin, a known Rgp inhibitor, and inhibition of Kgp with SBE was significantly higher than that with leupeptin at 500 μg/mL ($p < 0.05$). *P. gingivalis*-induced alveolar bone resorption was significantly suppressed by administration of SBE, with bone levels remaining comparable to non-infected animals ($p < 0.05$).

Conclusion: The present study suggests that SBE might be effective against *P. gingivalis*-associated alveolar bone resorption.

Yasushi Furuichi, DDS, PhD, Division of Periodontology and Endodontology, Department of Oral Rehabilitation, School of Dentistry, Health Sciences University of Hokkaido, 1757 Kanazawa Ishikari-Tobetsu, Hokkaido, Japan
Tel: +81 133 23 1211
Fax: +81 133 23 1414
e-mail: furuichi@hoku-iryo-u.ac.jp

Key words: alveolar bone resorption; *Porphyromonas gingivalis*; sword bean

Accepted for publication December 13, 2013

Periodontitis is a chronic inflammatory disease caused by periodontopathic bacteria such as *Porphyromonas gingivalis*, a gram-negative anaerobe, which has been implicated as the most important pathogen in chronic periodontitis due in part to its expression of potent virulence factors (1,2). Gingipain, which comprises Arg-gingipain (Rgp) and Lys-gingipain (Kgp) variants, is one such virulence factor, accounting for at least 85% of the cysteine protease activity of *P. gingivalis* (3).

Herbal remedies have been used in Eastern medicine for centuries, but the active ingredients of most medicinal herbs are still uncharacterized. In the context of their safety in daily use, herbal remedies have the great merit of having been used safely in Eastern medicine since ancient times. Recently, some herbal ingredients have attracted attention in modern medicine, such as tea catechin, which was reported to be effective in reducing probing pocket depth when applied as an adjunct to scaling and root planing (4). To the best of our knowledge, there are few other reports describing the use of medicinal herbs for the treatment of periodontitis.

Sword beans are domesticated plant species native to tropical Asia and Africa. The fruit has long been used in Chinese herbal medicine and in Japanese folk medicine for treating discharge of pus and for its anti-inflammatory properties, but are rarely consumed as a food because of the antinutritional factors they contain (e.g., protease inhibitors) (5). Canavanine is a non-protein amino acid naturally occurring in sword beans, and is a structural analogue of arginine. Canavanine inactivates an arginine-degrading enzyme of *Escherichia coli* (6), suggesting that Rgp might be inhibited by canavanine by a similar mechanism. In this study, we examined the possible application of sword bean extract (SBE) for the treatment of alveolar bone resorption associated with *P. gingivalis*. The effect of SBE on experimental *P. gingivalis*-induced alveolar bone resorption in rats was examined, as were its antibacterial effects and its cytotoxicity against human epithelial cells *in vitro*.

Material and methods

Preparation of sword bean extract

Sword beans were cultivated in Kagoshima, Japan and SBE was prepared by Yoshitome Co., Ltd. (Chiba, Japan). In brief, ground sword beans were blended in 50% ethanol. After filtration to remove the insoluble fraction, the remaining extract was lyophilized to give a powder. This powder was reconstituted in sterile distilled water for use in subsequent experiments.

High-performance liquid chromatography analysis

Chemicals— Solvents and chemicals were of HPLC grade. Water was passed through a Milli-Q water purification system (Millipore Corp., Bedford, MA, USA). Canavanine was obtained from Sigma-Aldrich Co., Ltd. (St Louis, MO, USA). Acetonitrile was obtained from Kanto Chemical Co., Ltd. (Tokyo, Japan). Dabsyl chloride reagent was obtained from Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan).

Derivatization with dabsyl chloride— Derivatization was performed according to the method described by Ekanayake *et al.* (7) with minor modifications. The SBE or canavanine standards were dissolved in 100 μL of reaction buffer (0.3 M $NaHCO_3$, pH 8.6) and mixed with 200 μL dabsyl chloride reagent (25 mg/mL acetone). The samples were incubated in a water bath at 70°C for 20 min with intermediate mixing at 1, 5 and 12 min, and then kept on ice for 8 min. Mobile phase A consisted of NaH_2PO_4 (9 mM), 4% dimethyl formamide and 0.2% triethylamine, pH adjusted to 6.55 with phosphoric acid. Nineteen microliter phase A was added to 1 μL of each sample and mixed thoroughly. The SBE and standards were then filtered through cotton on a Pasteur pipette and injected into the HPLC.

High-performance liquid chromatography analysis— An HPLC was coupled to an autosampler (Waters 2695 Alliance200; Waters Corp., Milford, MA, USA) and a ultraviolet-visible detector (Waters TM 996 Photodiode Array Detector; Waters Corp). Empower2 (Waters Corp.) software was used for system control, data collection and peak integration. The volume of the injection loop was 100 μL. The chromatographic conditions were adopted from Ekanayake *et al.* and further optimized (7). The column (Mightysil RP-18 GP; 5 μm; 250 × 4.6 mm i.d.; Kanto Chemical) equipped with a guard cartridge was maintained at 50°C. Mobile phase B consisted of aqueous acetonitrile (80% v/v). For the best resolution with a short elution time, a gradient system was used starting at 40% B for 15 min and then increased to 58% B from 15 to 17 min, followed by an increase to 100% B from 17 to 42 min. The flow rate was 1 mL/min, the injection volume was 20 μL and the eluate was monitored at 447 nm. Identification and quantification were achieved by comparing retention times and peak areas with those from a standard sample run under the same conditions.

Determination of minimum inhibitory concentration and minimum bactericidal concentration

Eight species were employed in the study: *Streptococcus mutans* Ingbritt, *Streptococcus sanguinis* ATCC 10556, *Streptococcus salivarius* ATCC 9222, *Lactobacillus casei* F. N1, *Actinomyces viscosus* ATCC 19246, *Aggregatibacter actinomycetemcomitans* ATCC 29522, *P. gingivalis* ATCC 33277, and *Fusobacterium nucleatum* JCM 6328. For minimum inhibitory concentration (MIC) testing, bacteria were grown under anaerobic conditions (N_2, 80%; H_2, 10%; CO_2, 10%) to stationary phase in 3% tryptic soy broth supplemented with 0.5% yeast extract, 5 mg/mL hemin, and 1 mg/mL menadione (TYHM broth) as appropriate to the species being tested. Thereafter, bacteria were

1600765, 2014, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jre.12166 by South China Agricultural, Wiley Online Library on [02/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

grown to stationary phase in TYHM broth then adjusted by dilution in broth to approximately 1×10^7 cells/mL (estimated spectrophotometrically). For each organism, the total microscopic count was first established relative to optical density at a wavelength of 600 nm. Strains were cultured anaerobically for 48 h at 37°C in TYHM broth. Concentrations of SBE, canavanine and chlorhexidine gluconate solution (CHX; Sigma-Aldrich) were prepared in broth. A negative control solution of broth containing no antimicrobial agent was also prepared. The broths were cultured anaerobically overnight at 37°C and examined for growth the following day. Tubes were examined for turbidity and the lowest concentration of test agent that fully inhibited the growth of organisms was recorded as the MIC. The minimum bactericidal concentration (MBC) determination was used to assess if the inhibitory effect observed in MIC determinations was through a lethal (bactericidal) action. Samples (100 μL) removed from MIC tubes that exhibited no turbidity were inoculated on to TYHM blood agar plates containing 5% sheep blood. The plates were subsequently incubated for 5 d and visible growth recorded. The MBC was the lowest concentration of test reagent in the tubes that killed all bacteria (8,9).

Cells and culture conditions

KB cells (ATCC CCL17) were maintained in Dulbecco's modified Eagle's medium (Sigma-Aldrich) supplemented with 10% heat-inactivated fetal bovine serum (Sigma-Aldrich), penicillin (100 U/mL) and streptomycin (100 μg/mL) at 37°C in 5% CO_2 in air.

Determination of cytotoxicity using trypan blue staining assay

KB cells were plated in 96-well plate at a concentration of 2×10^4 cells/well 24 h before the experiment. SBE, canavanine, CHX, leupeptin or cystatin was added to the wells and the plates were incubated for 1 min or 24 h. In the trypan blue staining assay, trypan blue (100 μL) was added to each well; unstained (viable) cells and stained (dead) cells were counted under a microscope using a hemacytometer. Cell viability (%) was determined by dividing the viable cell count by the total (viable + nonviable) cell count.

Effect of sword bean extract, canavanine, leupeptin and cystatin on Arg- and Lys-gingipain activities

The inhibitory activity of SBE, canavanine, leupeptin and cystatin on Rgp and Kgp was evaluated by colorimetric assay using synthetic substrates (10). In brief, *P. gingivalis* were harvested by centrifugation (10,000 *g* for 20 min), washed and suspended in 50 mM phosphate-buffered saline (pH 7.4) at 1×10^6 cells/mL. Benzoyl-arginine-*p*-nitroanilide (Sigma-Aldrich) and tosyl-glycine-proline-lysine-*p*-nitroanilide (Sigma-Aldrich), in 100 μL of 0.1 M Tris–HCl (pH 8.0) containing 1 mM dithiothreitol, were used as substrates (final concentration, 0.5 mM) for Rgp and Kgp, respectively. The substrates were dispensed into the wells of a 96-well microtiter plate. Bacterial cell suspension (50 μL) and different concentrations of SBE, canavanine, leupeptin or cystatin were added to the substrate and incubated at 37°C for 30 min. In the control wells, 50 μL of Tris–HCl was added in place of the bacterial suspension. Optical density at a wavelength of 405 nm (OD_{405}) was determined by the microtiter plate reader. Relative enzymatic activity was determined as follows: [(OD_{405} with bacterial cells and reagent − OD_{405} of control)/(OD_{405} with bacteria cells − OD_{405} of control)] × 100. Degradation obtained in the absence of the SBE, canavanine, leupeptin and cystatin was normalized to 100%.

Rat model of experimental *P. gingivalis*-induced alveolar bone resorption

Experimental alveolar bone resorption was induced in rats using a modification of the technique described by Nakajima *et al.* (11). Twenty male, specific-pathogen-free, 4-wk-old Wistar rats were obtained from a commercial farm (Hokudo Co., Ltd., Hokkaido, Japan) for use in these experiments. Because crowding or isolation may be stressful for experimental animals, rats were housed throughout the experiment in cages containing no more than three animals each. Rats were allowed free access to water and hard food briquettes and were maintained under a 12 h light/dark cycle (light on between 08:00 and 20:00 h) at a temperature of 22°C and relative humidity of 50%.

As shown in Fig. 1, rats were given sulfamethoxazole (1 mg/mL) and trimethoprim (200 μg/mL) in drinking water, *ad libitum*, for 4 d to reduce their commensal oral flora, followed by a 4 d antibiotic-free period before being challenged with *P. gingivalis*. Rats were orally challenged with *P. gingivalis* suspended in 5% carboxymethylcellulose (CMC), each rat receiving 0.5 mL (1×10^9 cells/mL) by oral gavage six times at 48 h intervals (11). To study the inhibition of alveolar bone loss, rats were randomly divided into four groups (A–D) of five animals. Group A was infected orally with *P. gingivalis*. Group B was orally administered with 2 mg of SBE suspended in 0.5 mL of CMC and then infected with *P. gingivalis* (in CMC) 10 min after administration of the SBE. Group C was administered with 0.5 mL of CMC alone, with no infection. Group D served as the negative control. At the end of the experimental period, all animals were killed by decapitation under diethyl ether anesthesia. The experimental procedures of this study were reviewed and approved by the Committee of Ethics on Animal Experiments at the Health Sciences University of Hokkaido and were carried out under the guidelines for animal experimentation at the Health Sciences University of Hokkaido.

Assessment of alveolar bone loss by digital histomorphometry

Using a digital camera system, the buccal surfaces of the maxillary and

16000765, 2014, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jre.12166 by South China Agricultural, Wiley Online Library on [02/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

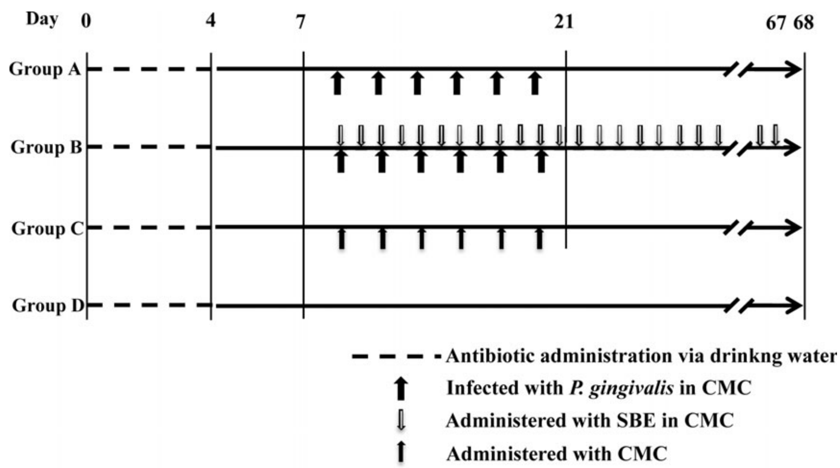

Fig. 1. Experimental procedure. Rats were divided into four experimental groups (n = 5 per group). Group A was orally infected with *P. gingivalis*; group B was orally administered with SBE in CMC and infected with *P. gingivalis*; group C was orally administered with CMC alone; and group D (negative control) was not infected with *P. gingivalis*. *P. gingivalis* were suspended in CMC when challenged. CMC, carboxymethylcellulose; SBE, sword bean extract.

mandibular posterior dentition were recorded in a standardized manner. The standardization of recording was assured using fixed reference points on the rat jaws. The line tool was used to make bone resorption measurements on all molars in each quadrant from the cemento–enamel junction (CEJ) to the alveolar bone crest (ABC). The surface perimeters of the CEJ and ABC were traced using Adobe Photoshop (Adobe Systems Co., Ltd., San Jose, CA, USA). The area of bone loss (in mm^2) was instantly imprinted over the digital image. The areas between the CEJ and ABC were calculated using ImageJ image analysis software (National Institutes of Health, Bethesda, MD, USA) (Fig. 2) (12).

Statistical analysis

Data were analyzed by one-way analysis of variance and Tukey's tests when indicated.

Results

High-performance liquid chromatography analysis

The chromatographic conditions employed by Ekanayake *et al.* (7) were initially used to run standard canavanine samples and then further optimized. A chromatogram obtained from the analysis of standard canavanine is shown in Fig. 3. The peak elution at 9 min corresponds to hydrolyzed excess dabsyl chloride reagent. The standard canavanine samples gave rise to two peaks at retention times of 20.6 min (arrow B) and 30.3 min (arrow C), indicated by arrows in Fig. 3. The ratio between the peak areas was similar in the concentration range used. The standard curve was based on the peak eluting at 30.3 min (arrow C) because this peak was better resolved. A linear relationship (R^2 = 0.99363) between peak area and canavanine concentration was observed over a range of 1–2000 nmol. A chromatogram obtained from the SBE is shown in Fig. 4. The peaks corresponding to canavanine are indicated by arrows. The peak at 30.3 min is large and well resolved, and the amount of canavanine in the SBE was calculated based on this peak area and found to be present at 6.4%.

Fig. 2. Morphometric evaluation of alveolar bone loss. Representative rat left jaw showing the buccal horizontal bone loss area by morphometry. The area outlined between cemento–enamel junction and alveolar bone crest in each tooth represents the area of horizontal alveolar bone resorption (in mm^2).

Effect of sword bean extract and canavanine on the growth of oral bacteria

Growth of *P. gingivalis* and *F. nucleatum* was inhibited by SBE with MIC values of 29.5 and 1875 μg/mL, respectively. Canavanine also inhibited the growth of *P. gingivalis* and *F. nucleatum*, with MIC values of 1458.5 and 1875 μg/mL, respectively (Table 1). The MBC of SBE for *P. gingivalis* and *F. nucleatum* was 29.5 and 1875 μg/mL, respectively. The MBC of canavanine for *P. gingivalis* and *F. nucleatum* was 1458.5 and 2083.5 μg/mL respectively (Table 2). However, the growth of *S. mutans*, *S. sanguinis*, *S. salivarius*, *L. casei*, *A. viscosus* and *A. actinomycetemcomitans* was not affected by either SBE or canavanine.

Cytotoxicity of sword bean extract, canavanine, chlorhexidine gluconate, leupeptin and cystatin on KB cells using trypan blue assay

At 31.25 μg/mL for 1 min, neither SBE, canavanine nor cystatin had a minimum toxic effect on KB cells *in vitro*. Cytotoxicity of SBE, canavanine and cystatin was less than that exerted by CHX and leupeptin even at higher concentrations (500 μg/mL

16000765, 2014, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jre.12166 by South China Agricultural, Wiley Online Library on [02/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

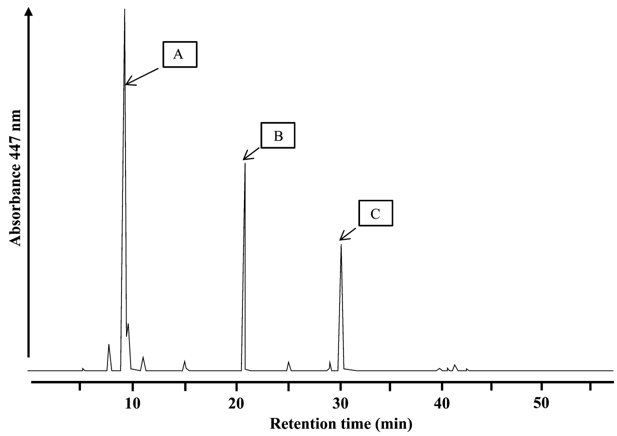

Fig. 3. High-performance liquid chromatography chromatogram of a standard canavanine sample. Standard canavanine was processed by high-performance liquid chromatography as described in Material and methods. Eluate was analyzed at 447 nm. The peaks corresponding to excess hydrolyzed dabsyl chloride (A) and canavanine (B and C) are indicated by arrows.

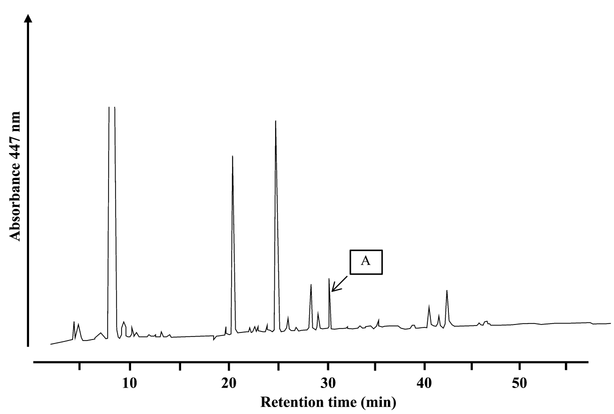

Fig. 4. High-performance liquid chromatography chromatogram of a sword bean extract sample. Sword bean extract was processed by high-performance liquid chromatography as described in Material and methods. Eluate was analyzed at 447 nm. The peaks corresponding to excess canavanine are indicated by arrows (A).

for 1 min) (Fig. 5A). Similarly, after 24 h incubation with a concentration up to 500 μg/mL, cytotoxic effects of SBE, canavanine and cystatin on KB cells were low. In contrast, CHX was markedly cytotoxic at concentrations of 31.25 μg/mL or more for 24 h (Fig. 5B).

Effect of sword bean extract, canavanine, leupeptin and cystatin on *P. gingivalis* protease activity

The effects of SBE, leupeptin and cystatin on Rgp and Kgp activities are shown in Fig. 6. SBE and leupeptin significantly inhibited Rgp activity at 0.0305–500 and 0.488–500 μg/mL, respectively. The inhibitory effect of leupeptin on Rgp was higher than that of SBE at 1.95–125 μg/mL, but the same at 500 μg/mL (Fig. 6A). Cystatin did not have any inhibitory effect on Rgp at any concentration (Fig. 6A).

SBE and leupeptin significantly inhibited the Kgp activity at 0.0305–500 and 0.488–500 μg/mL, respectively. The inhibitory effect of SBE on Kgp was significantly higher than that of leupeptin at 0.0305–0.122 and 500 μg/mL (Fig. 6B). Cystatin did not

Table 1. Minimum inhibitory concentration (μg/mL) of reagents on various microorganisms

	P. gingivalis		*F. nucleatum*		*S. mutans*		S. salivalius		S. sanguins		*L. casei*		*A. viscosus*		*A. actinomycetemcomitans*	
Sample size	Average	SD	Average	SD	Average	SD	Average	SD	Average	SD	Average	SD	Average	SD	Average	SD
SBE	59	0.003	1875	1.083	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
CAV	1094	0.955	1875	1.083	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
CHX	39	0	39	0	39	0	39	0	39	0	39	0	39	0	39	0

SBE, sword bean extract; CAV, canavanine; CHX, chlorhexidine gluconate solution; NA, no activity at the highest concentration tested.

16000765, 2014, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jre.12166 by South China Agricultural, Wiley Online Library on [02/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Minimum bactericidal concentration (μg/mL) of reagents on various microorganisms

	P. gingivalis		*F. nucleatum*		*S. mutans*		S. salivalius		S. sanguins		*L. casei*		*A. viscosus*		*A. actinomycetemcomitans*	
Sample size	Average	SD	Average	SD	Average	SD	Average	SD	Average	SD	Average	SD	Average	SD	Average	SD
SBE	59	0.003	1875	1.083	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
CAV	1094	0.955	2083.5	0.723	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
CHX	39	0	39	0	39	0	39	0	39	0	39	0	39	0	39	0

SBE, sword bean extract; CAV, canavanine; CHX, chlorhexidine gluconate solution; NA, no activity at the highest concentration tested.

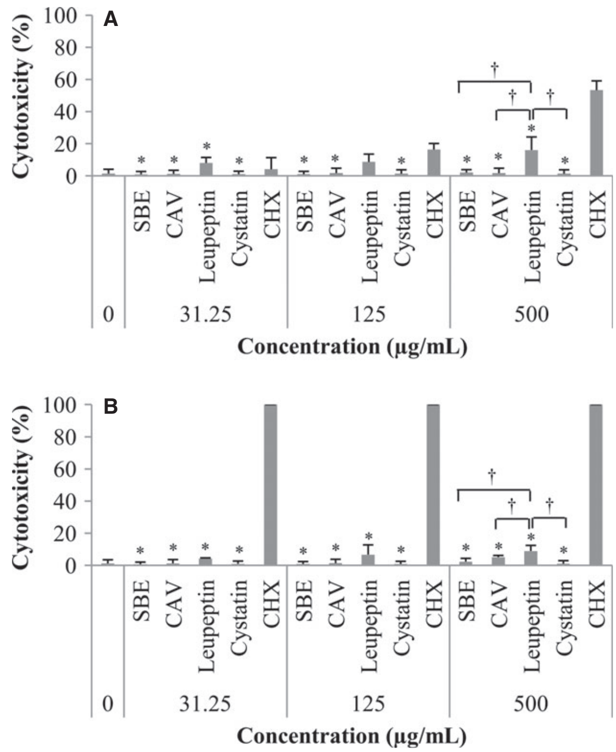

Fig. 5. Trypan blue test for cytotoxicity effects against human oral epithelial cells. Human oral epithelial cells (KB cells) were prepared as described in Material and methods. For the experiments, reagents were added to each well for 1 min (A) or 24 h (B). Viable and total cell counts were made using a hemocytometer, and viable cells expressed as a percentage of the total cell count as a measure of cytotoxicity. *Statistically significant difference ($p < 0.05$) compared to CHX; †Statistically significant difference ($p < 0.05$) compared with leupeptin as determined by ANOVA with Tukey's test. CAV, canavanine; CHX, chlorhexidine gluconate; SBE, sword bean extract.

have any inhibitory effect on Kgp at any concentration (Fig. 6B). Rgp and Kgp activities were not significantly ($p < 0.05$) inhibited by canavanine at concentrations < 2 mg/mL (data not shown).

Effect of sword bean extract on *P. gingivalis*-induced alveolar bone resorption

As shown in Fig. 7, *P. gingivalis*-infected rats (group A) had significantly higher alveolar bone loss in the upper molars compared with non-infected groups (groups C and D). The alveolar bone loss in the upper molars was not different between groups C and D, indicating that administration of CMC did not affect alveolar bone resorption. Administration of SBE (group B) suppressed resorption around the upper molars to levels comparable to those observed in the non-infected groups (Fig. 7). A similar suppressive effect of SBE on *P. gingivalis*-induced alveolar bone resorption was observed in the lower jaw (data not shown).

Discussion

This is the first study to evaluate the antimicrobial effects of SBE and canavanine on oral bacteria and their cytotoxic effects on oral epithelial cells. SBE and canavanine inhibited the growth of *P. gingivalis* and *F. nucleatum*, but were not effective against other oral bacteria, including *S. mutans*, *S. sanguinis*, *S. salivarius*, *L. casei*, *A. viscosus* and *A. actinomycetemcomitans*. Although the observed antimicrobial effects were limited to gram-negative bacteria and significantly smaller than those of CHX, the MICs

16000765, 2014, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jre.12166 by South China Agricultural, Wiley Online Library on [02/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

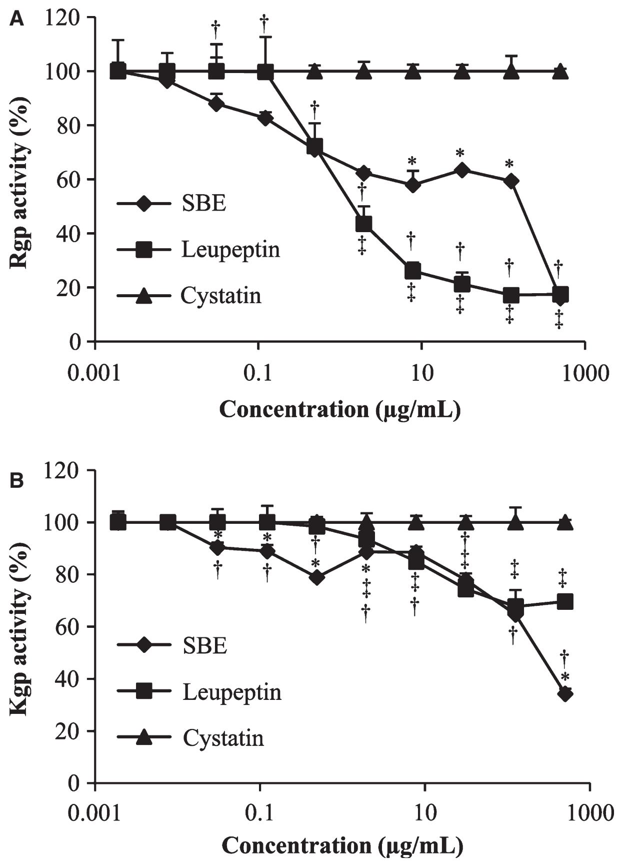

Fig. 6. Effects of SBE, leupeptin and cystatin on Rgp (A) and Kgp (B) of *P. gingivalis*. *P. gingivalis* were incubated with synthetic Rgp and Kgp substrates in the presence or absence of SBE, leupeptin or cystatin. Degradation of these substrates triggered a change on optical density at 405 nm and was indicative of Rgp and Kgp activity. Degradation in the absence of the abovementioned substances was normalized as 100%. Statistical significance of differences between the results in the presence and absence of SBE, leupeptin or cystatin were calculated using the test. *Statistically significant ($p < 0.05$) difference between SBE and leupeptin. †Statistically significant ($p < 0.05$) difference between SBE and cystatin. ‡Statistically significant ($p < 0.05$) difference between leupeptin and cystatin. Kgp, Lys-gingipain; Rgp, Arg-gingipain; SBE, sword bean extract.

of SBE and canavanine were lower than the cytotoxic concentration on the KB cell lines. This indicates that at effective antimicrobial concentrations, SBE is as safe for daily use as CHX mouth rinse. No plant extract product has been systematically evaluated in *in vitro* studies or assessed for long-term clinical effects except Listerine (13,14). However, Listerine is a mixture of four essential oils and has contained relatively high concentrations of alcohol (10–26%). An important characteristic of a mouth rinse for wide use is that it contain little to no alcohol, because dry mouth is becoming a serious problem among the elderly, in part because of multiple medication use (15). For this reason, alcohol-free mouth rinses containing active hydrophilic compounds such as SBE or canavanine are preferable.

Inhibition of *P. gingivalis* growth by SBE may be due in part to suppression of its proteolytic activities, as reported by Kadowaki *et al.* (15,16). In their study, growth of *P. gingivalis* was suppressed when gingipain inhibitors (KYT-1 and/or KYT-36) were added to the growth media. This finding suggests that suppressing proteolytic enzyme activity deprives *P. gingivalis* of essential nutrition needed for growth.

In this study, antiproteolytic activities of SBE were observed at the concentration below MIC. *P. gingivalis* proteases such as Rgp and Kgp are involved in the pathogenesis of periodontal disease by: (i) directly degrading periodontal structural proteins; (ii) disrupting host defense mechanisms; (iii) activating or stimulating the expression of hemagglutinin; (iv) processing and translocating adhesion molecules; and (v) inducing or stimulating inflammation through the production of chemical mediators (17,18). Kadowaki *et al.* (16) reported that gingipain inhibitors reduced the pathogenicity of *P. gingivalis*. The activity of Rgp and Kgp is found to be attributable to either Arg-X- or Lys-X-specific cysteine proteinase (16). Canavanine has structural similarity with arginine, suggesting that canavanine might interfere with Arg-X activity in Rgp (19–21). In this study, SBE and canavanine inhibited not only Rgp activity, but also Kgp activity. However, canavanine did not show inhibitory activity against Rgp or Kgp at lower concentration relative to SBE or leupeptin, suggesting that SBE has additional properties not present in canavanine that allow it to suppress both Rgp and Kgp activity. Both leupeptin and cystatin are cysteine protease inhibitors. As leupeptin inhibited Rgp (22), but cystatin did not suppress *P. gingivalis* gingipain (23), we used leupeptin and cystatin as positive and negative controls to inhibit gingipain. At 500 µg/mL, inhibition of Rgp with SBE was comparable to that with leupeptin, and the inhibition of Kgp with SBE was significantly higher than that with leupeptin. In addition, suppression of Rgp and Kgp activity with SBE was, though partial, pronounced more than with leupeptin at low concentration (0.0305–1.95 µg/mL), suggesting that SBE even at a diluted concentration *in vivo* might suppress both Rgp and Kgp activities effectively. To the best of our knowledge, protease inhibitors such as EDTA and *N*-tosyl-*L*-lysyl chloromethyl ketone cannot be applied in the prevention and treatment of periodontitis because they are toxic to oral epithelial cells at the concentrations required to inhibit Rgp and Kgp activity (22,24–26). The inhibitory effects of leupeptin on Rgp

16000765, 2014, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jre.12166 by South China Agricultural, Wiley Online Library on [02/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

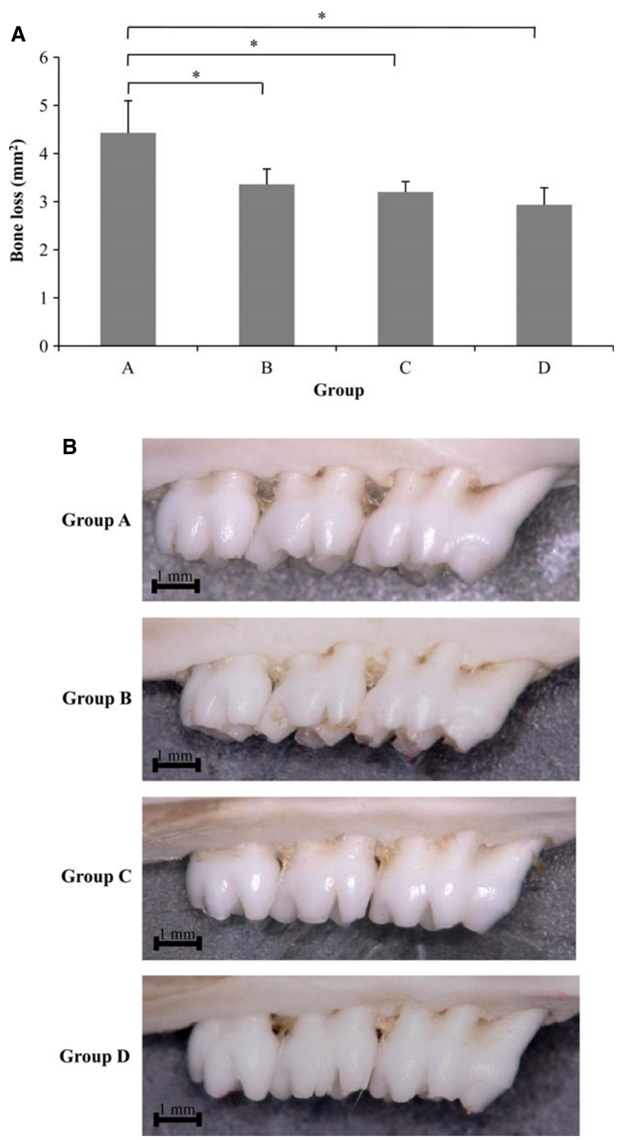

Fig. 7. Effects of sword bean extract on *P. gingivalis*-induced alveolar bone resorption. (A) Morphometric quantification of bone loss. (B) Mean (SD) of alveolar bone loss in the upper arch. Rats were infected (or not) with oral *P. gingivalis* for several days before the measurement of bone levels, which was performed by comparing the area from the cemento–enamel junction to the alveolar bone crest around three molars on the left side of the upper jaw. (A) *P. gingivalis*-infected rat; (B) *P. gingivalis*-infected rat treated with sword bean extract; (C) carboxymethylcellulose rat; (D) control rat. *Statistically significant ($p < 0.05$) difference between groups.

activities and in a rat model were demonstrated by Kitano *et al*. (22). In their animal experiments, leupeptin was effective in reducing experimental gingivitis at a concentration of 1 mg/mL. However, cytotoxicity of leupeptin on KB cells was significantly higher than that of SBE at 500 μg/mL in the present study. In addition, SBE at a concentration of 500 μg/mL showed bactericidal activity, was less cytotoxic and inhibited Rgp and Kgp. All these findings together indicated that SBE has a sufficiently low risk of cytotoxicity to be deemed safe to use *in vivo* for treating *P. gingivalis*-associated periodontitis. The efficacy of SBE in reducing periodontal bone loss suggests that its action of suppressing these enzymes might have therapeutic relevance in *P. gingivalis*-associated alveolar bone resorption. SBE suppressed alveolar bone resorption in rats orally infected with *P. gingivalis* in the present study. The relatively weak antimicrobial activity of SBE against *P. gingivalis* suggests that this is not the only activity involved in its effects on bone resorption. Its suppressive actions against Rgp and Kgp activities could be an additional component in this mechanism. Although several reports have suggested that canavanine is an important component of sword beans (7), we found that canavanine was not as effective as SBE against periodontopathic bacteria and gingipains. No *in vitro* or *in vivo* studies have evaluated the effects of both crude and purified components of medicinal herbs on periodontopathic bacteria. Several researchers reported that biologically active polyphenols purified from green tea extracts were effective against periodontopathic bacteria (27); however, studies reporting the antimicrobial effects of crude green tea extracts could not be found in the literature. The MICs of tea polyphenols against *P. gingivalis* were reported as 250–1000 μg/mL (27), indicating that the inhibitory effects of canavanine on periodontopathic bacteria (Table 1) are similar to those of tea polyphenols.

In Japanese folk medicine, crude extracts were often thought to be more effective than purified substances (28). The reason for this is not clear, but one explanation is that combinations of small amounts of chemicals may be more effective than a single chemical. Rodent experiments and human epidemiological studies provide evidence that green tea can inhibit tumor development and growth. However, the concentrations of tea polyphenols needed to suppress tumors *in vitro* are considerably higher than concentrations that can be readily obtained *in vivo* (29). The synergistic effects of epigallocatechin gallate combined with (–)-epicatechin against cancer were reported by Suganuma *et al.* (30), suggesting that small amounts of other components (i.e., epicatechin) augmented the antitumor effects of epigallocatechin gallate. Further studies are needed to elucidate any active

1600765, 2014, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jre.12166 by South China Agricultural, Wiley Online Library on [02/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

components of SBE and determine the clinical efficacy of SBE on the initiation and progression of periodontitis.

Acknowledgements

The authors thank Dr. Masaru Terasaki for technical support and Dr. Sonoko Masuda for valuable advice on analytical issues (Division of Pharmaceutical Health Sciences, Faculty of Pharmaceutical Sciences, Health Sciences University of Hokkaido).

References

1. Holt SC, Kesavalu L, Walker S, Genco CA. Virulence factors of *Porphyromonas gingivalis*. *Periodontol 2000* 1999;**20**: 168–238.
2. Lamont RJ, Jenkinson HF. Life below the gum line: pathogenic mechanisms of *Porphyromonas gingivalis*. *Microbiol Mol Biol Rev* 1998;**62**:1244–1263.
3. DeCarlo AA, Paramaesvaran M, Yun PL, Collyer C, Hunter N. Porphyrin-mediated binding to hemoglobin by the HA2 domain of cysteine proteinases (gingipains) and hemagglutinins from the periodontal pathogen *Porphyromonas gingivalis*. *J Bacteriol* 1999;**181**:3784–3791.
4. Hirasawa M, Takada K, Makimura M, Otake S. Improvement of periodontal status by green tea catechin using a local delivery system: a clinical pilot study. *J Periodontal Res* 2002;**37**:433–438.
5. Laurena AC, Revilleza MJR, Mendoza EMT. Polyphenols, phytate, cyanogenic glycosides, and trypsin inhibitor activity of several Philippine indigenous food legumes. *J Food Compost Anal* 1994;**7**: 194–202.
6. Li L, Li Z, Chen D *et al*. Inactivation of microbial arginine deiminases by L-canavanine. *J Am Chem Soc* 2008;**130**: 1918–1931.
7. Ekanayake S, Skog K, Asp NG. Canavanine content in sword beans (*Canavalia gladiata*): analysis and effect of processing. *Food Chem Toxicol* 2007;**45**:797–803.
8. Mann CM, Cox SD, Markham JL. The outer membrane of *Pseudomonas aeruginosa* NCTC 6749 contributes to its tolerance to the essential oil of *Melaleuca alternifolia* (tea tree oil). *Lett Appl Microbiol* 2000;**30**:294–297.
9. Percival RS, Devine DA, Duggal MS, Chartron S, Marsh PD. The effect of cocoa polyphenols on the growth, metabolism, and biofilm formation by *Streptococcus mutans* and *Streptococcus sanguinis*. *Eur J Oral Sci* 2006;**114**:343–348.
10. Yamanaka A, Kouchi T, Kasai K, Kato T, Ishihara K, Okuda K. Inhibitory effect of cranberry polyphenol on biofilm formation and cysteine proteases of *Porphyromonas gingivalis*. *J Periodont Res* 2007;**42**:589–592.
11. Nakajima K, Hamada N, Takahashi Y *et al*. Restraint stress enhances alveolar bone loss in an experimental rat model. *J Periodont Res* 2006;**41**:527–534.
12. Verma RK, Rajapakse S, Meka A *et al*. *Porphyromonas gingivalis* and *Treponema denticola* mixed microbial infection in a rat model of periodontal disease. *Interdiscip Perspect Infect Dis* 2010;**2010**:605125.
13. Drake D, Villhauer AL. An in vitro comparative study determining bactericidal activity of stabilized chlorine dioxide and other oral rinses. *J Clin Dent* 2011;**22**:1–5.
14. Gordon JM, Lamster IB, Seiger MC. Efficacy of Listerine antiseptic in inhibiting the development of plaque and gingivitis. *J Clin Periodontol* 1985;**12**:697–704.
15. Guggenheimer J, Moore PA. Xerostomia: etiology, recognition and treatment. *J Am Dent Assoc* 2003;**134**:61–69 quiz 118–119.
16. Kadowaki T, Baba A, Abe N *et al*. Suppression of pathogenicity of *Porphyromonas gingivalis* by newly developed gingipain inhibitors. *Mol Pharmacol* 2004;**66**:1599–1606.
17. Imamura T, Pike RN, Potempa J, Travis J. Pathogenesis of periodontitis: a major arginine-specific cysteine proteinase from *Porphyromonas gingivalis* induces vascular permeability enhancement through activation of the kallikrein/kinin pathway. *J Clin Invest* 1994;**94**:361–367.
18. Okamoto K, Nakayama K, Kadowaki T, Abe N, Ratnayake DB, Yamamoto K. Involvement of a lysine-specific cysteine proteinase in hemoglobin adsorption and heme accumulation by *Porphyromonas gingivalis*. *J Biol Chem* 1998;**273**:21225–21231.
19. Teas HJ. Effect of canavanine on mutants of Neurospora and *Bacillus subtilis*. *J Biol Chem* 1951;**190**:369–375.
20. Walker JB. Canavanine and homoarginine as antimetabolites of arginine and lysine in yeast and algae. *J Biol Chem* 1955;**212**:207–215.
21. Wheeler MA, Smith SD, Garcia-Cardena G, Nathan CF, Weiss RM, Sessa WC. Bacterial infection induces nitric oxide synthase in human neutrophils. *J Clin Invest* 1997;**99**:110–116.
22. Kitano S, Irimura K, Sasaki T *et al*. Suppression of gingival inflammation induced by *Porphyromonas gingivalis* in rats by leupeptin. *Jpn J Pharmacol* 2001;**85**:84–91.
23. Abrahamson M, Wikstrom M, Potempa J, Renvert S, Hall A. Modification of cystatin C activity by bacterial proteinases and neutrophil elastase in periodontitis. *Mol Pathol* 1997;**50**:291–297.
24. Higuchi M, Singh S, Chan H, Aggarwal BB. Protease inhibitors differentially regulate tumor necrosis factor-induced apoptosis, nuclear factor-kappa B activation, cytotoxicity, and differentiation. *Blood* 1995;**86**:2248–2256.
25. Johansson A, Kalfas S. Characterization of the proteinase-dependent cytotoxicity of *Porphyromonas gingivalis*. *Eur J Oral Sci* 1998;**106**:863–871.
26. Park H, Ko MY, Paik MK, Soh CT, Seo JH, Im KI. Cytotoxicity of a cysteine proteinase of adult *Clonorchis sinensis*. *Korean J Parasitol* 1995;**33**:211–218.
27. Sakanaka S, Aizawa M, Kim M, Yamamoto T. Inhibitory effects of green tea polyphenols on growth and cellular adherence of an oral bacterium, *Porphyromonas gingivalis*. *Biosci Biotechnol Biochem* 1996;**60**:745–749.
28. Shimizu M, Weinstein IB. Modulation of signal transduction by tea catechins and related phytochemicals. *Mutat Res* 2005; **591**:147–160.
29. Chow HH, Cai Y, Hakim IA *et al*. Pharmacokinetics and safety of green tea polyphenols after multiple-dose administration of epigallocatechin gallate and polyphenon E in healthy individuals. *Clin Cancer Res* 2003;**9**:3312–3319.
30. Suganuma M, Okabe S, Kai Y, Sueoka N, Sueoka E, Fujiki H. Synergistic effects of (–)-epigallocatechin gallate with (–)-epicatechin, sulindac, or tamoxifen on cancer-preventive activity in the human lung cancer cell line PC-9. *Cancer Res* 1999;**59**:44–47.

16000765, 2014, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jre.12166 by South China Agricultural, Wiley Online Library on [02/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License